Title: Assignment 1

Group 3:

Member 1: Tamam Fajar Briliansyah (5025231142)

Member 2: Dapunta Adyapaksi Ratyanasja (5025231187)

Case Information

Basic administrative details that identify the case and this report (NIST SP 800-86, Reporting).

Task	1
Case ID / Reference number	DF-A1-G3-2026
Report version and date	v1.0, 5 October 2026
Examiner(s)	Dapunta Adyapaksi Ratyanasja (5025231187)
Requesting party	Course lecturer, Digital Forensics (T), Department of Informatics, Institut Teknologi Sepuluh Nopember
Case type	Disk forensics: USB flash drive imaging and deleted-file recovery
Date and location of examination	5 October 2026, Surabaya, Indonesia

Task	2
Case ID / Reference number	FOREN-A1-G3-T2
Report version and date	v1.0, 5 October 2026
Examiner(s)	Tamam Fajar Briliansyah (5025231142)
Requesting party	Course lecturer, Digital Forensics (T), Department of Informatics, Institut Teknologi Sepuluh Nopember
Case type	Image forensics (copy-move forgery detection)
Date and location of examination	5 to 6 October 2026, Surabaya, Indonesia

Introduction

Contains a brief description of the case study being solved.

Task 1: Digital Forensics & File Recovery

Task 1 involves a 16 GB SanDisk Cruzer Switch USB flash drive. The USB drive was prepared as a clean test device formatted with FAT32. A known JPEG file was copied onto it and then permanently deleted using the Shift + Delete, and a physical forensic image was then acquired with FTK Imager. The deleted file was recovered from the forensic image using TestDisk and validated against the retained original using SHA-256 hashing. The workflow was designed to preserve the physical state of the media following deletion, perform recovery from the forensic image (rather than directly from the USB device), and maintain integrity records for both the forensic image and the recovered file.

Task 2: Image Copy-Move Forgery Detection (Groups 1–7)

Digital images are widely used as evidence, and they can be altered with little visible trace. One common alteration is copy-move forgery, in which a region of an image is copied and pasted into another location of the same image, either to hide an object or to duplicate one. This case examines whether the Copy-Move Forgery tool of the open-source toolkit Sherloq, used with the BRISK keypoint detector, can detect such forgeries, and how sensitive the detection is to the tool parameters. Ten forged images from the CoMoFoD benchmark dataset were analysed, and the detections were compared against the ground-truth masks published with the dataset.

Scope and Objectives

Task 1:

The investigation addressed the following questions:

- Can a deleted JPEG file still be identified and recovered from a physical forensic image of a FAT32 USB flash drive?
- Can the recovered file be shown to be identical to the original test file using a cryptographic hash?
- Can the integrity of the forensic image be demonstrated before and after examination?

Out of scope were malware analysis, copy-move or splicing analysis, recovery of unrelated historical data, and repair of the USB file system. The USB was group-owned training media and the activity was performed solely for the authorized course assignment.

Task 2:

The examination answers two questions. First, can the Copy-Move Forgery tool of Sherloq with the BRISK detector locate the cloned regions in the test images? Second, how do the tool parameters (Response, Matching, Distance, Cluster) and the characteristics of the forgery (transformation type and region size) affect detection?
In scope are ten forged images from the CoMoFoD small dataset (version 2), analysed only with the Copy-Move Forgery tool using the BRISK detector. Out of scope are the ORB and AKAZE

detectors, splicing detection, deep learning based detection, tests on unmodified images, and the post-processed variants of the images (JPEG compression, noise, blur, colour reduction). The examination was carried out on a public research dataset, so no special authorization was required. It is assumed that the ground-truth masks supplied with the dataset are correct.

Tools Used

Contains the various tools used in the investigation process. Record the exact version of each tool so that the results can be reproduced and, where available, note whether the tool has been independently tested (e.g., in the NIST Computer Forensics Tool Testing (CFTT) program).

Task 1:

No	Tool	Version	Source / URL	Purpose	Validation notes
1	Microsoft DiskPart	10.0.26100.1150	Windows built-in utility	Remove prior Ventoy layout, initialize MBR, create one FAT32 partition	Result verified with detail disk, list volume, and Windows Properties
2	FTK Imager	Product 8.3.0.27 File 8.2.2.999	Exterro FTK Imager	Physical acquisition to Raw (dd) image and acquisition verification	MD5 and SHA-1 verification matched. No bad blocks reported
3	TestDisk	7.2	TestDisk 7.2	Read FAT32 forensic image, locate deleted directory entry, recover the deleted JPEG	Recovered file validated against the original using SHA-256
4	PowerShell Get-FileHash	Built-in cmdlet (version not separately recorded)	Microsoft Windows PowerShell	Compute SHA-256 for original, USB copy, forensic image, and recovered file	Independent hash comparisons returned identical values

File description	FTK Imager
Type	Application
File version	8.2.2.999
Product name	Exterro® FTK® Imager
Product version	8.3.0.27

File description	TestDisk
Type	Application
File version	7.2.0.0
Product name	TestDisk
Product version	7.2

Task 2:

No	Tool	Version	Source / URL	Purpose	Validation notes
1	Sherloq	0.94f (window title), Git commit 3fe95fc (2026-07-16), branch master	https://github.com/GuidoBartoli/sherloq	Image forensic analysis, Copy-Move Forgery module	Open-source academic tool, not evaluated in NIST CFTT. Results validated against dataset ground truth
2	Python (CPython, Windows x86-64)	3.11.15	https://www.python.org	Runtime for Sherloq	General-purpose runtime, installed through uv
3	OpenCV (opencv-contrib-python-headless)	4.11.0.86	https://opencv.org	BRISK keypoint detection and matching used by Sherloq	Library used inside Sherloq
4	PySide6	6.7.3	https://doc.qt.io/qtforpython-6/	Graphical interface of Sherloq	Library used inside Sherloq
5	uv	0.11.23	https://docs.astral.sh/uv/	Virtual environment and dependency installation	Package manager
6	GNU sha256sum	coreutils (Linux shell environment)	https://www.gnu.org/software/coreutils/	SHA-256 hashing at acquisition and first verification (5 October 2026)	Standard algorithm, output identical to PowerShell Get-FileHash
7	PowerShell Get-FileHash	Built-in cmdlet (version not separately recorded)	Microsoft Windows PowerShell	Second SHA-256 verification after all Sherloq runs (6 October 2026)	All ten values matched the acquisition hashes

Description of Digital Evidence

Task 1:

The principal physical evidence was a SanDisk Cruzer Switch USB flash drive marketed as 16 GB. Before the exercise it contained a Ventoy boot configuration with a large exFAT partition and a 32 MB VTOYEFI partition. The device was intentionally sanitized for the controlled exercise, repartitioned using MBR, and formatted as a single FAT32 volume labelled

DF_GROUP3. A known JPEG test file was retained on the workstation as ground truth, copied to the USB, verified, and then permanently deleted.

Task 2:

The evidence consists of ten forged images (file suffix _F) selected from the CoMoFoD small dataset v2, published by the Faculty of Electrical Engineering and Computing, University of Zagreb (https://www.vcl.fer.hr/comofod/). Each image is a 512 x 512 pixel PNG in which one or more regions have been copied and pasted within the same image. For every image the dataset also provides a binary ground-truth mask (suffix _B) that marks both the source region and the pasted region. The ten sets cover each transformation category and region sizes from 403 to 28,405 pixels.

Evidence Inventory

Task 1

Item ID	Description	Make / model / OS version	Serial no. / image name	Source and condition	Size
E-01	Physical USB flash drive	SanDisk Cruzer Switch FAT32, MBR	Windows Disk ID: B65DE0C2	Group-owned training media; previously Ventoy; cleaned and reformatted; functional	15,744,892,928 bytes physical image size (~14.66 GiB)
E-02	Raw forensic image of E-01	Raw (dd) generated by FTK Imager	Group3_SanDisk16GB_20261005.001	Physical acquisition of E-01 after permanent deletion; stored read-only by procedure during analysis	15,744,892,928 bytes
E-03	Reference and target JPEG file	JPEG image	evidence_group3.jpg	Original retained in 01_Original; copy placed on E-01, deleted, then recovered to 03_Recovery	61,320 bytes

Task 2

Item ID	Description	Make / model / OS version	Serial no. / IMEI / image name	Source and condition	Size
E-01	Forged image, translation, region 2,122 px	Not applicable (dataset image)	001_F.png	CoMoFoD small v2, unaltered	429.4 KB
E-02	Forged image, translation, region 9,852 px	Not applicable	005_F.png	CoMoFoD small v2, unaltered	418.8 KB
E-03	Forged image, translation, region 403 px	Not applicable	010_F.png	CoMoFoD small v2, unaltered	371.8 KB

E-04	Forged image, translation, region 28,405 px	Not applicable	024_F.png	CoMoFoD small v2, unaltered	452.2 KB
E-05	Forged image, rotation R2, region 14,186 px	Not applicable	050_F.png	CoMoFoD small v2, unaltered	538.1 KB
E-06	Forged image, rotation R180, region 2,632 px	Not applicable	063_F.png	CoMoFoD small v2, unaltered	543.3 KB
E-07	Forged image, scaling S101, region 9,852 px	Not applicable	085_F.png	CoMoFoD small v2, unaltered	395.2 KB
E-08	Forged image, scaling S40, region 15,478 px	Not applicable	103_F.png	CoMoFoD small v2, unaltered	536.5 KB
E-09	Forged image, vertical distortion V115, region 14,186 px	Not applicable	126_F.png	CoMoFoD small v2, unaltered	535.3 KB
E-10	Forged image, combination S80 R45, region 3,540 px	Not applicable	168_F.png	CoMoFoD small v2, unaltered	338.6 KB

The ground-truth masks (gt/NNN_B.png) are reference material. Their hashes are listed in Appendix B.

Integrity Verification (Hash Values)

Record cryptographic hash values (e.g., SHA-256) computed at acquisition and again before analysis to show that the evidence has not been altered.

Task 1

Item ID	Algorithm	Hash at acquisition / reference	Hash at verification	Match
E-02	MD5	aa7022c0ba9d08e1d977b84036ba49f1	aa7022c0ba9d08e1d977b84036ba49f1	Yes
E-02	SHA-1	25fa3b9f34f842a9ea0dbc39fdd28c9a2415c3c3	25fa3b9f34f842a9ea0dbc39fdd28c9a2415c3c3	Yes
E-02	SHA-256	C2E427E0A413D0D6A43C9ED750C16FB2A9177ED9784F7D8211DCA699D8EB24A1	C2E427E0A413D0D6A43C9ED750C16FB2A9177ED9784F7D8211DCA699D8EB24A1	Yes

E-03	SHA-256	458A79BCB8303CB8A077D26699552D3E33434FEBCA0E4293E7F1547D68F4D591	458A79BCB8303CB8A077D26699552D3E33434FEBCA0E4293E7F1547D68F4D591	Yes

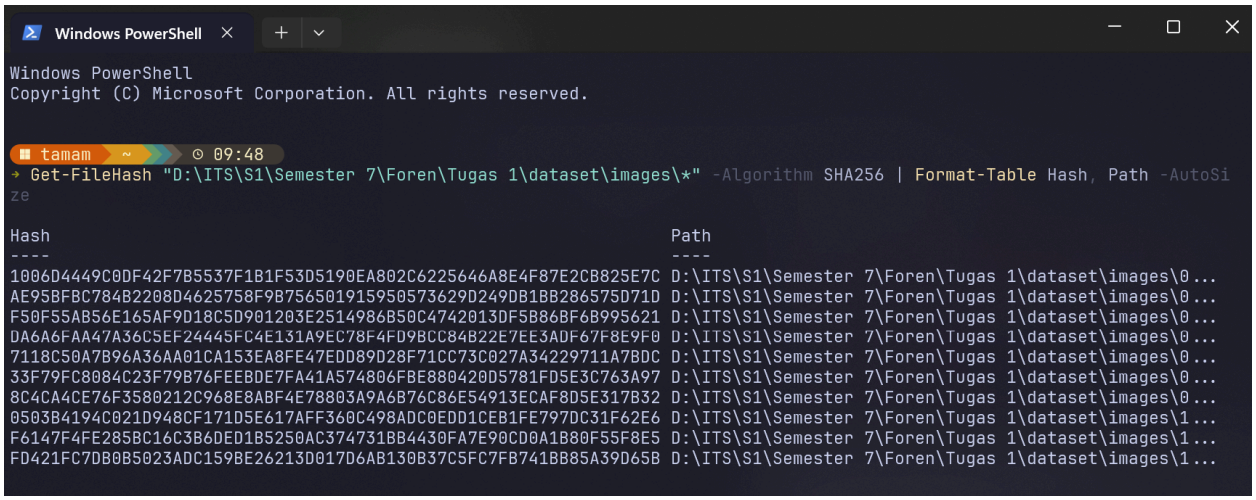
Integrity was checked at multiple points. FTK Imager performed built-in verification using MD5 and SHA-1. SHA-256 was additionally computed for the raw image after acquisition and again after TestDisk examination. The original and recovered JPEG files were also compared using SHA-256.

Task 2

Item ID	Algorithm	Hash at acquisition	Hash at verification	Match (Yes/No)
E-01	SHA-256	1006d4449c0df42f7b5537f1b1f53d5190ea802c6225646a8e4f87e2cb825e7c	1006d4449c0df42f7b5537f1b1f53d5190ea802c6225646a8e4f87e2cb825e7c	Yes
E-02	SHA-256	ae95bfbc784b2208d4625758f9b756501915950573629d249db1bb286575d71d	ae95bfbc784b2208d4625758f9b756501915950573629d249db1bb286575d71d	Yes
E-03	SHA-256	f50f55ab56e165af9d18c5d901203e2514986b50c4742013df5b86bf6b995621	f50f55ab56e165af9d18c5d901203e2514986b50c4742013df5b86bf6b995621	Yes
E-04	SHA-256	da6a6faa47a36c5ef24445fc4e131a9ec78f4fd9bcc84b22e7ee3adf67f8e9f0	da6a6faa47a36c5ef24445fc4e131a9ec78f4fd9bcc84b22e7ee3adf67f8e9f0	Yes
E-05	SHA-256	7118c50a7b96a36aa01ca153ea8fe47edd89d28f71cc73c027a34229711a7bdc	7118c50a7b96a36aa01ca153ea8fe47edd89d28f71cc73c027a34229711a7bdc	Yes
E-06	SHA-256	33f79fc8084c23f79b76feebde7fa41a574806fbe880420d5781fd5e3c763a97	33f79fc8084c23f79b76feebde7fa41a574806fbe880420d5781fd5e3c763a97	Yes
E-07	SHA-256	8c4ca4ce76f3580212c968e8abf4e78803a9a6b76c86e54913ecaf8d5e317b32	8c4ca4ce76f3580212c968e8abf4e78803a9a6b76c86e54913ecaf8d5e317b32	Yes
E-08	SHA-256	0503b4194c021d948cf171d5e617aff360c498adc0edd1ceb1fe797dc31f62e6	0503b4194c021d948cf171d5e617aff360c498adc0edd1ceb1fe797dc31f62e6	Yes
E-09	SHA-256	f6147f4fe285bc16c3b6ded1b5250ac374731bb4430fa7e90cd0a1b80f55f8e5	f6147f4fe285bc16c3b6ded1b5250ac374731bb4430fa7e90cd0a1b80f55f8e5	Yes

E-10	SHA-256	fd421fc7db0b5023adc159be26213d017d6ab130b37c5fc7fb741bb85a39d65b	fd421fc7db0b5023adc159be26213d017d6ab130b37c5fc7fb741bb85a39d65b	Yes

Hashes were computed on 5 October 2026 at 15:59 and verified for all 30 files (ten images, ten masks, ten originals) at 16:23. After all Sherloq runs, the ten forged images were hashed again with PowerShell Get-FileHash on 6 October 2026 at 09:48 (Figure 3), and all ten values matched the acquisition hashes.



Chain of Custody

Document who handled the evidence, when, and why, from the moment it was obtained until the end of the investigation

Task 1:

Date / time	Item ID	Released by	Received by	Purpose	Storage location
05-Oct-2026	E-01	Group 3 training media	Group 3 examiners	Preparation of controlled USB evidence and placement/deletion of test file	Examiner-controlled USB device
05-Oct-2026 after deletion	E-01	Group 3 examiners	Group 3 examiners	Physical acquisition with FTK Imager; no intentional file-system writes after deletion	USB retained and later ejected/stored
05-Oct-2026 after acquisition	E-02	FTK Imager acquisition process	Group 3 examiners	Examination and deleted-file recovery from working forensic image	02_Acquisition\Group3_SanDisk16GB_20261005.001
05-Oct-2026 after recovery	E-03	TestDisk recovery output	Group 3 examiners	Hash and visual comparison against original ground-truth file	03_Recovery\evidence_group3.jpg

Task 2:

Date / time	Item ID	Released by	Received by	Purpose	Storage location
5 October 2026, 15:56	E-01 to E-10	CoMoFoD website (University of Zagreb)	Tamam Fajar Briliansyah	Download of the dataset archive CoMoFoD_small_v2	Examiner workstation, folder comofod_small
5 October 2026, 15:59	E-01 to E-10	Folder comofod_small	Folder dataset	Selection of ten sets, copy of forged images, masks and originals, computation of SHA-256 hashes	Examiner workstation, Tugas 1\dataset
5 October 2026, 16:23	E-01 to E-10	Folder dataset	Folder dataset	First hash verification, all 30 files matched	Examiner workstation, Tugas 1\dataset
5 October 2026, 18:40 to 19:11, and 6 October 2026, 09:25 to 09:27	E-01 to E-10	Folder dataset	Sherloq (read-only analysis)	Copy-move analysis	Examiner workstation, Tugas 1\dataset
6 October 2026, 09:48	E-01 to E-10	Folder dataset	Folder dataset	Second hash verification of the ten forged images after the experiments, all matched	Examiner workstation, Tugas 1\dataset

Forensic Process Performed

Task 1:

1. Preparation

Step 1.1. Identify and sanitize the USB media.

The SanDisk Cruzer Switch had previously been configured as a Ventoy boot device. DiskPart showed two removable volumes: a 14 GB exFAT Ventoy partition and a 32 MB VTOYEFI partition. After confirming that Disk 1 was the USB device, the command clean all was executed to overwrite the addressable sectors and remove the previous layout before the controlled experiment.

```
diskpart
list disk
```

```
select disk 1
detail disk
clean all
```

```
SanDisk Cruzer Switch USB Device
Disk ID: C07562BE
Type   : USB
Status : Online
Path   : 0
Target : 0
LUN ID : 0
Location Path : UNAVAILABLE
Current Read-only State : No
Read-only  : No
Boot Disk  : No
Pagefile Disk  : No
Hibernation File Disk  : No
Crashdump Disk  : No
Clustered Disk  : No

  Volume ###  Ltr  Label        Fs     Type        Size     Status     Info
  ----------  ---  -----------  -----  ----------  -------  ---------  --------
  Volume 4     D   Ventoy       exFAT  Removable     14 GB  Healthy
  Volume 5     E   VTOYEFI      FAT    Removable     32 MB  Healthy

DISKPART> clean all

DiskPart succeeded in cleaning the disk.
```

Step 1.2. Create a single FAT32 test volume.

The cleaned USB was initialized with an MBR partition table, one primary partition was created, and the volume was quick-formatted as FAT32 with the label DF_GROUP3. DiskPart then showed a single healthy removable FAT32 volume.

```
convert mbr
create partition primary
format fs=fat32 quick label=DF_GROUP3
assign
detail disk
list volume
```

```
DISKPART> detail disk

SanDisk Cruzer Switch USB Device
Disk ID: B65DE0C2
Type   : USB
Status : Online
Path   : 0
Target : 0
LUN ID : 0
Location Path : UNAVAILABLE
Current Read-only State : No
Read-only  : No
Boot Disk  : No
Pagefile Disk  : No
Hibernation File Disk  : No
Crashdump Disk  : No
Clustered Disk  : No

  Volume ###  Ltr  Label        Fs     Type        Size     Status     Info
  ----------  ---  -----------  -----  ----------  -------  ---------  --------
* Volume 4     E   DF_GROUP3    FAT32  Removable     14 GB  Healthy

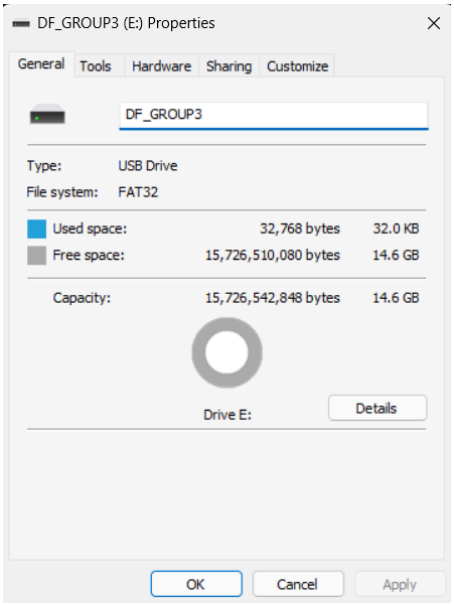
DISKPART> list volume

  Volume ###  Ltr  Label        Fs     Type        Size     Status     Info
  ----------  ---  -----------  -----  ----------  -------  ---------  --------
  Volume 0     C   OS           NTFS   Partition    952 GB  Healthy    Boot
  Volume 1         SYSTEM       FAT32  Partition    260 MB  Healthy    System
  Volume 2                      NTFS   Partition   1146 MB  Healthy    Hidden
  Volume 3         MYASUS       FAT32  Partition    260 MB  Healthy    Hidden
* Volume 4     E   DF_GROUP3    FAT32  Removable     14 GB  Healthy

DISKPART>
```

Step 1.3. Verify the prepared volume in Windows.

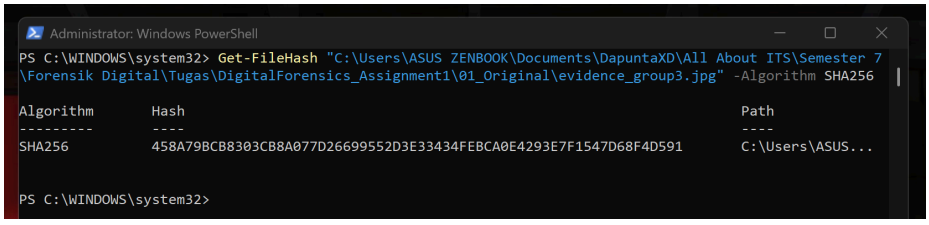
Windows Properties reported the DF_GROUP3 volume as FAT32 with a capacity of 15,726,542,848 bytes (14.6 GB usable file-system capacity). This confirmed that the previous Ventoy layout had been replaced by the intended single FAT32 volume.



2. Collection (Identification, Acquisition, and Preservation)

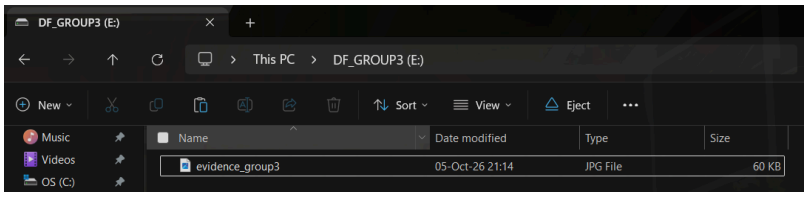
Step 2.1. Establish the original test file and SHA-256 baseline.

A distinctive JPEG named evidence_group3.jpg was retained in the workstation folder 01_Original as the reference file. SHA-256 was calculated before the file was copied to the USB. The resulting reference hash was 458A79BCB8303CB8A077D26699552D3E33434FEBCA0E4293E7F1547D68F4D591.



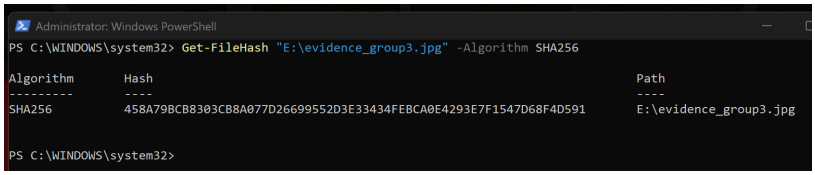
Step 2.2. Copy the test file to the USB.

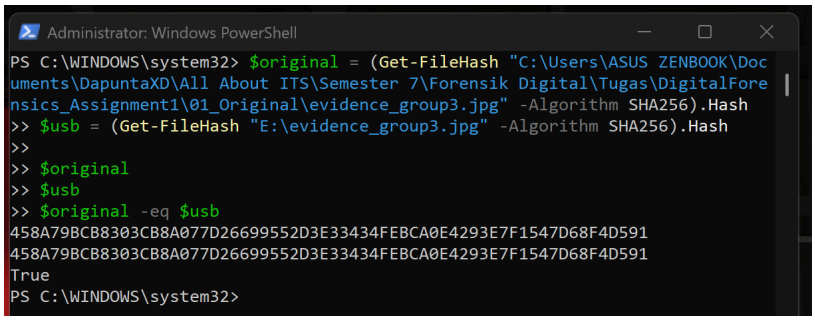
The JPEG was copied to the root of DF_GROUP3 (E:). File Explorer showed evidence_group3.jpg on the USB before deletion.



Step 2.3. Verify the USB copy using SHA-256.

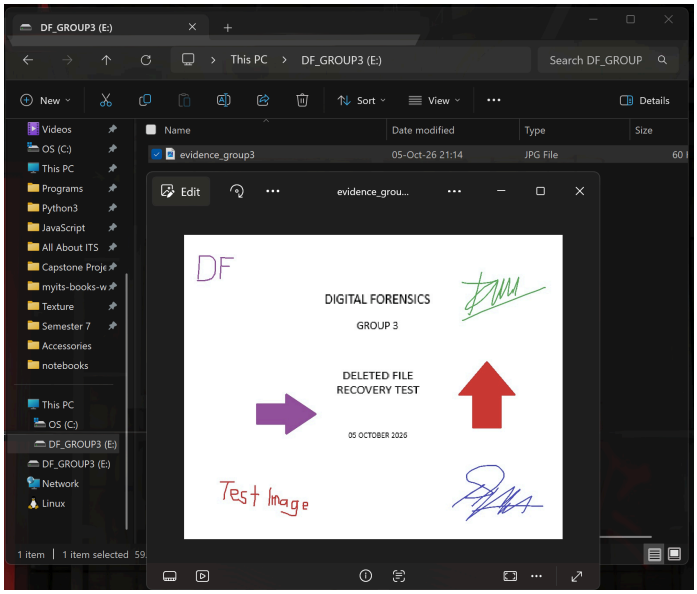
SHA-256 of E:\evidence_group3.jpg was calculated and matched the original reference hash. A direct PowerShell comparison returned True, confirming that the USB copy was identical to the retained original before deletion.





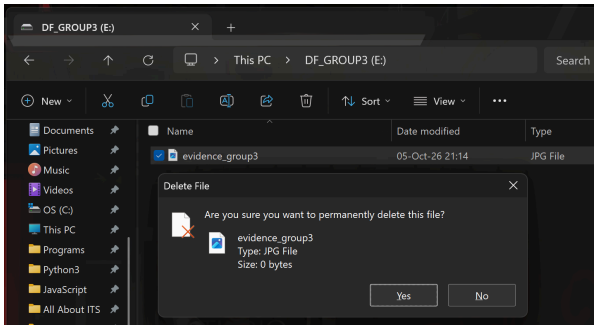
Step 2.4. Confirm the file is readable before deletion.

The JPEG was opened directly from the USB and displayed correctly. This provided a visual baseline that the file was intact and accessible before the deletion operation.



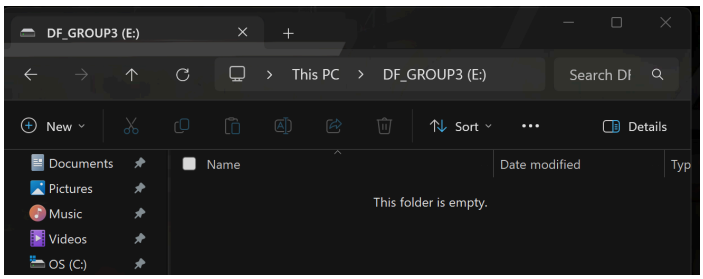
Step 2.5. Permanently delete the file.

The file was selected in File Explorer and deleted using Shift + Delete. The Windows confirmation dialog explicitly asked whether the file should be permanently deleted. The deletion was confirmed, bypassing the Recycle Bin.



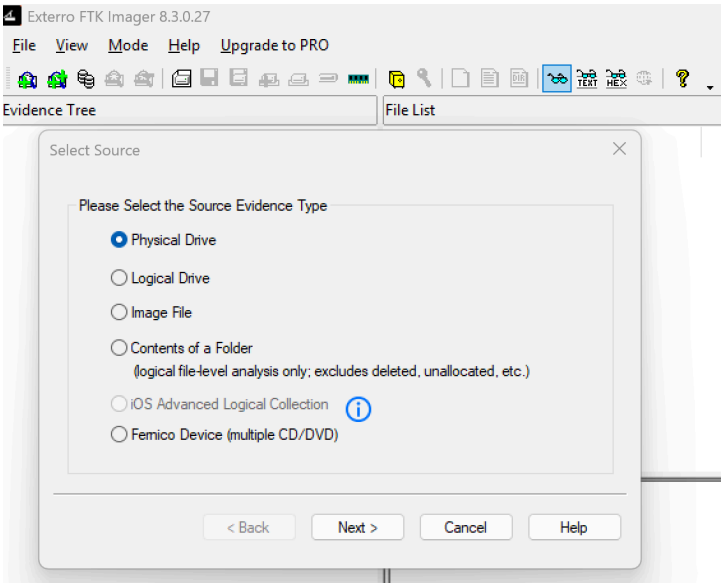
Step 2.6. Confirm the visible directory is empty.

After deletion, the root of DF_GROUP3 appeared empty in File Explorer. From this point onward, no intentional files or folders were written to the USB. The subsequent recovery work was performed against the forensic image.



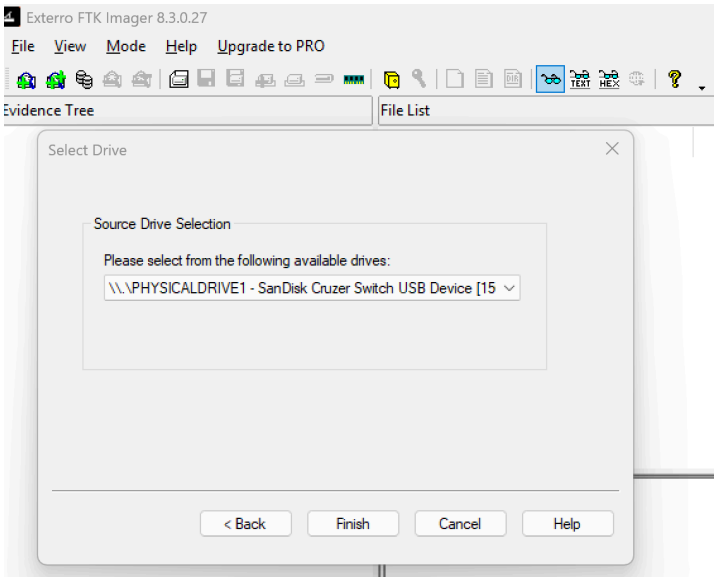
Step 2.7. Select physical acquisition in FTK Imager.

FTK Imager was started with administrative privileges. Create Disk Image was used and Physical Drive was selected as the evidence source so that the entire USB device, including file-system structures and unallocated space, would be acquired



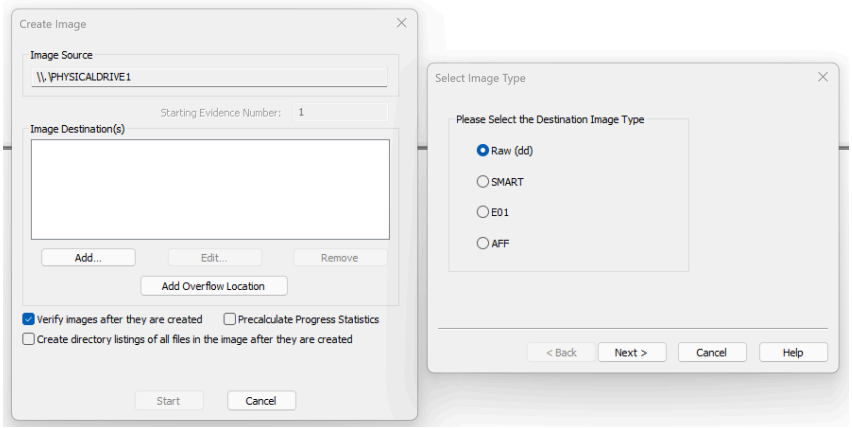
Step 2.8. Identify the correct physical source.

FTK Imager identified \.\PHYSICALDRIVE1 as the SanDisk Cruzer Switch USB device. This was selected as the acquisition source, consistent with the earlier DiskPart identification.



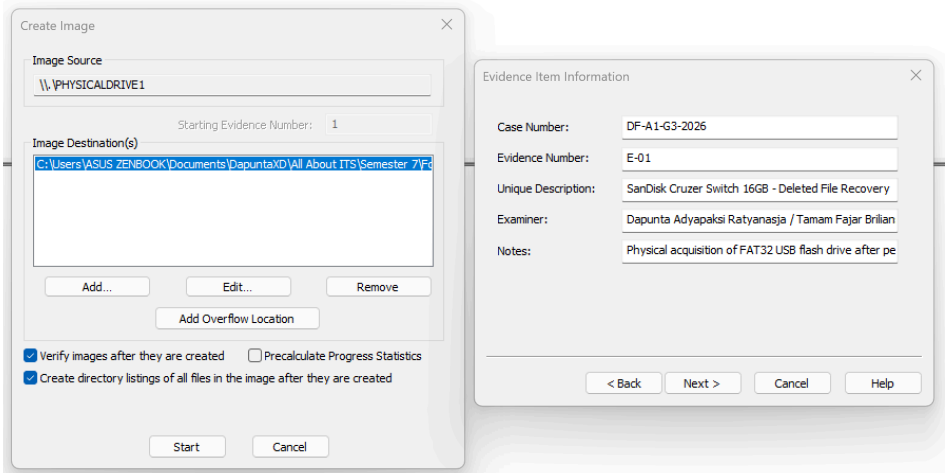
Step 2.9. Select Raw (dd) image format.

Raw (dd) was selected as the destination image type. This format preserves a sector-by-sector representation of the physical device and is broadly compatible with forensic tools such as TestDisk.



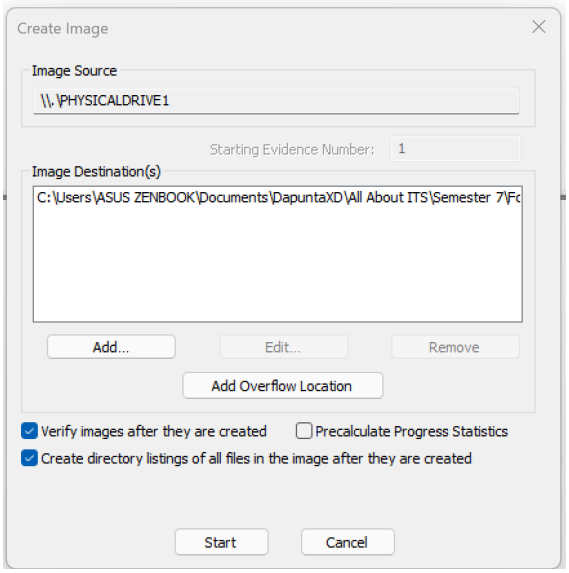
Step 2.10. Record evidence metadata.

The acquisition metadata recorded Case Number DF-A1-G3-2026, Evidence Number E-01, a description of the SanDisk device, the two group examiners, and a note that the acquisition occurred after permanent deletion of the target file.



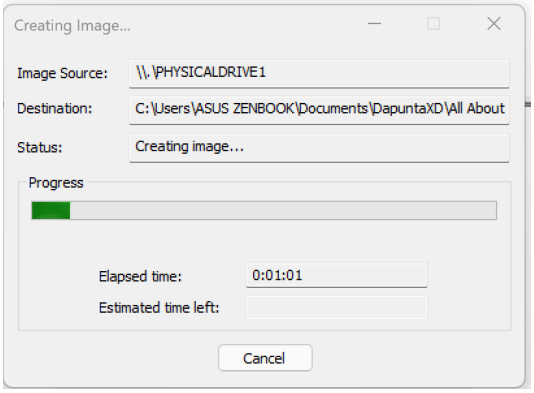
Step 2.11. Configure acquisition verification.

The image destination was set to the 02_Acquisition folder. FTK Imager was configured to verify the image after creation and to create a directory listing. The image base name was Group3_SanDisk16GB_20261005.



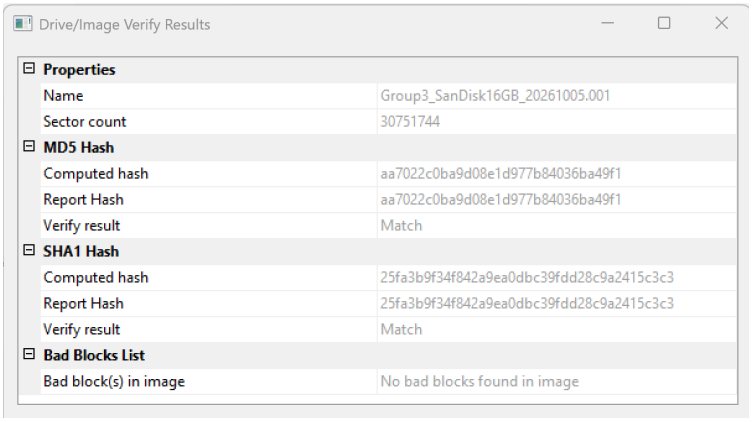
Step 2.12. Create the physical forensic image.

FTK Imager read the entire physical source and wrote the raw image to the workstation. No recovery actions were performed on the USB during acquisition.



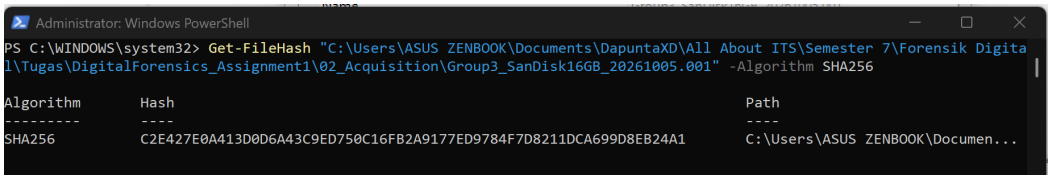
Step 2.13. Verify the forensic image.

FTK Imager completed built-in verification. The computed and report MD5 hashes matched, the computed and report SHA-1 hashes matched, and the application reported that no bad blocks were found. The raw image contained 30,751,744 sectors.



Step 2.14. Record a SHA-256 hash for the forensic image.

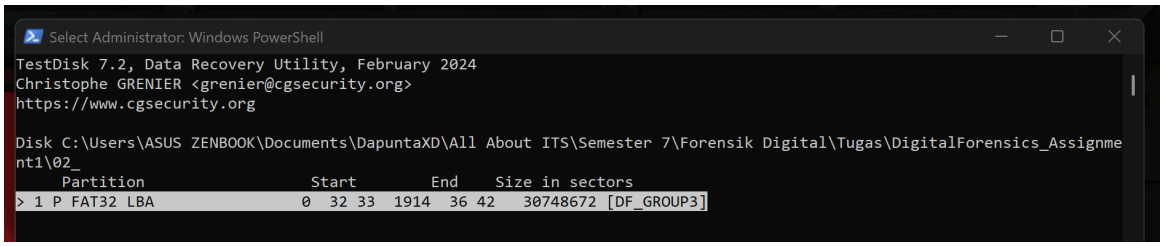
A SHA-256 hash was computed for Group3_SanDisk16GB_20261005.001 after acquisition. The recorded value was C2E427E0A413D0D6A43C9ED750C16FB2A9177ED9784F7D8211DCA699D8EB24A1. The physical USB was then no longer required for examination and could be retained separately.



3. Examination

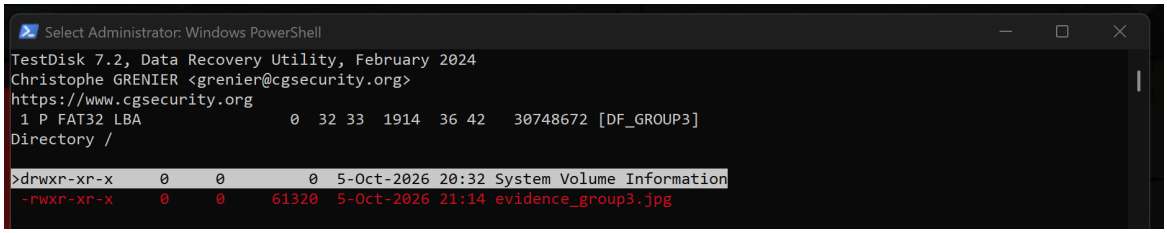
Step 3.1. Open the raw image in TestDisk and identify the FAT32 partition.

TestDisk 7.2 was executed with Group3_SanDisk16GB_20261005.001 as the input image. The MBR/Intel partition layout was used and the Advanced menu identified one FAT32 LBA partition labelled DF_GROUP3 with 30,748,672 sectors.



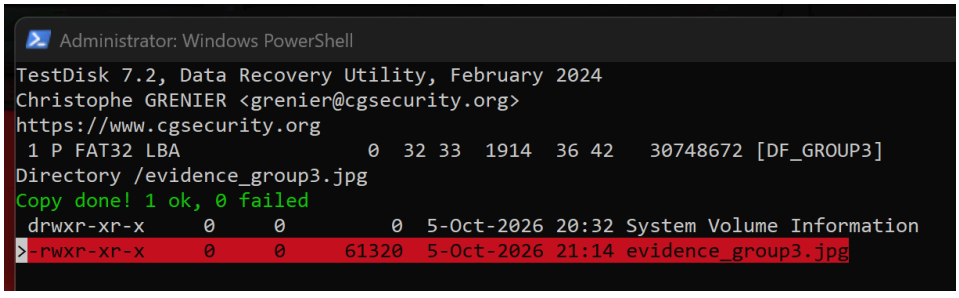
Step 3.2. Locate the deleted directory entry.

The Undelete function was used on the FAT32 partition. TestDisk displayed evidence_group3.jpg in red as a deleted entry with a size of 61,320 bytes and a timestamp of 05-Oct-2026 21:14. This confirmed that the deleted file metadata and content references remained recoverable in the forensic image.



Step 3.3. Recover the deleted file to a separate folder.

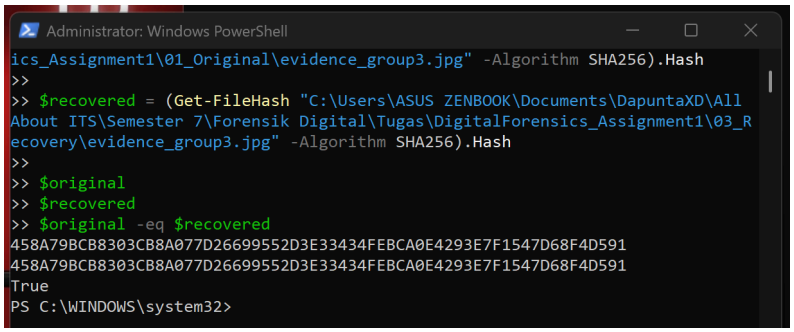
The deleted JPEG was copied from the forensic image to the workstation folder 03_Recovery. TestDisk reported Copy done! 1 ok, 0 failed. The recovery destination was outside both the physical USB and the forensic image.



4. Analysis

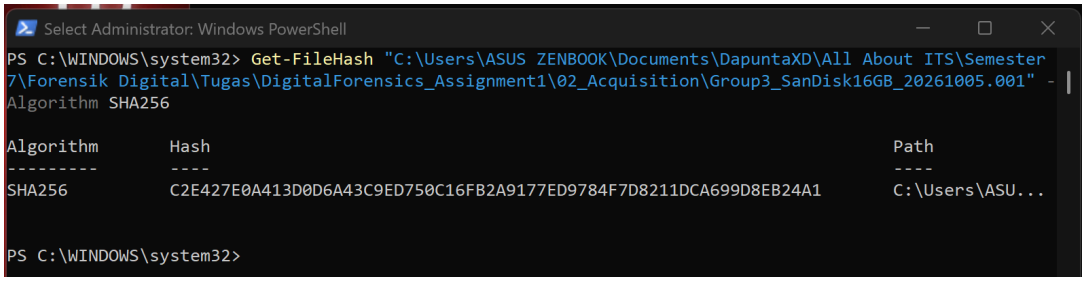
Step 4.1. Compare the recovered file with the original.

SHA-256 was calculated for the recovered JPEG and compared with the retained original file. Both values were 458A79BCB8303CB8A077D26699552D3E33434FEBCA0E4293E7F1547D68F4D591, and PowerShell returned True. Therefore, the recovered file was bit-for-bit identical to the original test file.



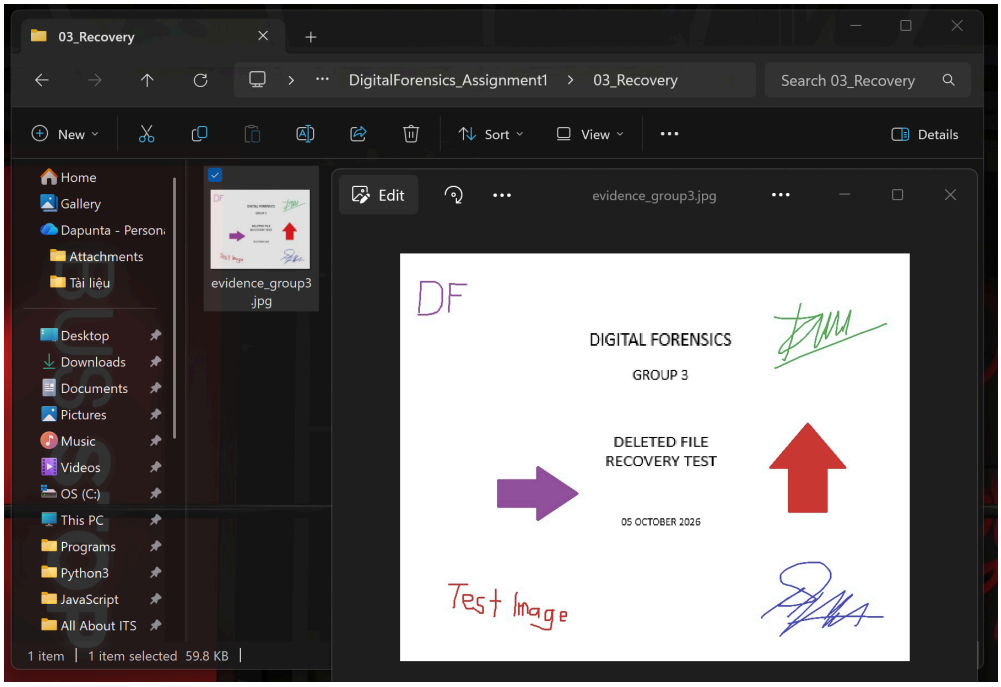
Step 4.2. Re-verify the forensic image after examination.

After TestDisk was closed, SHA-256 of the raw image was calculated again. The post-analysis value remained C2E427E0A413D0D6A43C9ED750C16FB2A9177ED9784F7D8211DCA699D8EB24A1, matching the value recorded immediately after acquisition. This demonstrates that the forensic image was not modified during examination.



Step 4.3. Perform visual validation of the recovered content.

The recovered evidence_group3.jpg was opened from 03_Recovery and displayed the same distinctive content as the pre-deletion file, including the DIGITAL FORENSICS and GROUP 3 text, colored arrows, date, and handwritten marks. This visual check supported the cryptographic verification.



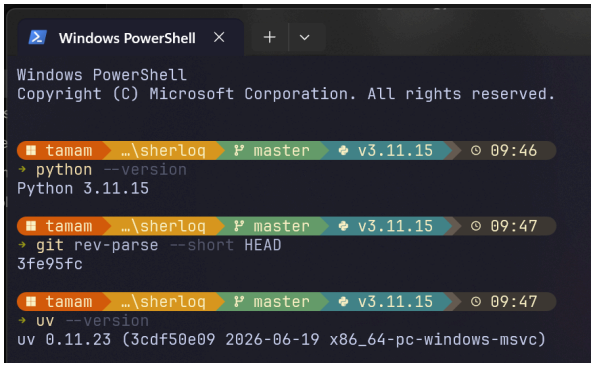
5. Reporting

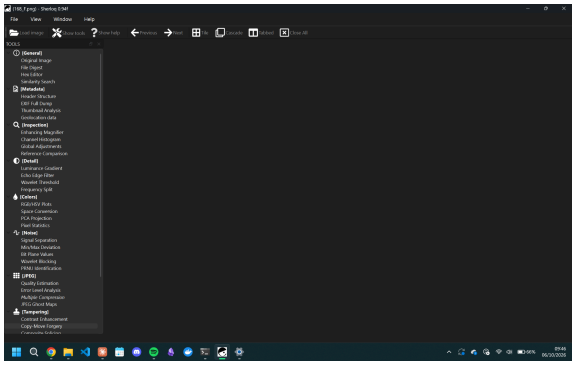
The examination was documented using screenshots captured at each material step, hash values recorded before and after recovery, FTK Imager verification results, and TestDisk recovery output. The report records the tool versions, source and destination paths, image format, integrity checks, findings, and limitations so that another examiner can reproduce the workflow with equivalent media and tools.

Task 2:

1. Preparation

The examination was performed on a personal 64-bit Windows workstation. Sherloq was cloned from its GitHub repository (commit 3fe95fc, Figure 1) and installed in an isolated Python 3.11.15 virtual environment created with uv 0.11.23, using gui/requirements.txt (OpenCV contrib headless 4.11.0.86, PySide6 6.7.3). The package python-magic-bin 0.4.14 was added because the Windows build of python-magic could not locate libmagic. The Sherloq main window, which reports version 0.94f, is shown in Figure 2. All analysis was done offline on copies of the dataset. The dataset is public and no legal or policy restrictions apply.





2. Collection (Identification, Acquisition, and Preservation)

The archive CoMoFoD_small_v2 (200 image sets, 512 x 512 pixels) was downloaded manually from the official dataset page. Ten sets (001, 005, 010, 024, 050, 063, 085, 103, 126, 168) were selected so that the sample covers four translation cases of different region sizes, two rotation cases, two scaling cases, one distortion case and one combination case. For each set the forged image and the ground-truth mask were copied into a dedicated dataset folder, leaving the downloaded archive untouched. The matching original images were copied as well but were not used in the analysis, so their hashes are not listed in Appendix B. The ground-truth masks were converted to clean 8-bit black-and-white PNG files. SHA-256 hashes of all 30 files were recorded with GNU sha256sum on 5 October 2026 at 15:59 and verified at 16:23, with every hash matching. After the experiments the ten forged images were verified again with PowerShell Get-FileHash (Figure 3). All analysis was done on the copies.

3. Examination

Each forged image was opened in Sherloq. The Copy-Move Forgery tool detects keypoints with the selected detector, matches keypoints with similar descriptors within the same image, discards matches that are too close together, groups the remaining matches into clusters, and draws lines between matched points. Only the BRISK detector was used. Four parameters control the result: Response (the share of keypoints kept, ranked by strength), Matching (how different two descriptors may be and still count as a match), Distance (the minimum spacing between matched points, as a percentage of the image size) and Cluster (the minimum number of matches in a

group).
Three BRISK parameter sets were applied. C1 is the Sherloq default (Response 90%, Matching 20%, Distance 15%, Cluster 5) and was run on all ten forged images as an unbiased baseline. C2 is a single fixed relaxation (Response 100%, Matching 25%, Distance 15%, Cluster 3) applied to all ten forged images. C3 is an image-specific set, chosen with knowledge of the ground truth, so it indicates the best achievable result rather than a realistic forensic scenario, and it was not run for E-06. Candidate parameters were chosen in advance with a Python re-implementation of the Sherloq copy-move pipeline, which was run over a range of Matching, Distance, Cluster and Response values on the same images and compared with the ground-truth masks. The parameters of every run are listed in Table 1. In each run the status line (Keypoints, Filtered, Matches, Clusters, Regions) and a screenshot were recorded. The ground-truth mask was viewed separately and was not loaded through the Load mask button, because that button restricts keypoint search to the masked area.

4. Analysis

For each run, the result was compared with the ground-truth mask, which marks both the source region and the pasted region. A run was classified as Detected when at least one cluster was found and the match lines connect the two marked regions without notable extra lines; Partial when the marked regions were connected but additional lines appear elsewhere; Not detected when no cluster was found; and False detection when clusters were found but the lines do not connect the marked regions. The classification was made by visual comparison of each screenshot with the mask and is therefore a judgment of the examiners. Table 1 lists every run with the values reported by Sherloq and the resulting classification.

Table 1. Run Log

Item	Config	Response	Matching	Distance	Cluster	Keypoints	Filtered	Matches	Clusters	Regions	Outcome	Screenshot
E-01	C1	90	20	15	5	2643	2078	26	6	1	Detected	E01_C1
E-01	C2	100	25	15	3	2643	2643	124	54	2	Partial	E01_C2
E-01	C3	90	20	15	3	2643	2078	26	10	1	Detected	E01_C3
E-02	C1	90	20	15	5	1596	1588	568	485	1	Detected	E02_C1
E-02	C2	100	25	15	3	1596	1595	750	711	1	Detected	E02_C2

E-02	C3	90	20	15	3	1596	1588	568	523	1	Detected	E02_C3
E-03	C1	90	20	15	5	1589	1166	12	0	0	Not detected	E03_C1
E-03	C2	100	25	15	3	1589	1589	108	41	1	False detection	E03_C2
E-03	C3	100	30	15	3	1589	1589	246	121	2	False detection	E03_C3
E-04	C1	90	20	15	5	2239	1757	192	86	1	Detected	E04_C1
E-04	C2	100	25	15	3	2239	2238	510	267	6	Partial	E04_C2
E-04	C3	100	25	15	5	2239	2238	510	298	6	Partial	E04_C3
E-05	C1	90	20	15	5	1517	506	22	8	1	Detected	E05_C1
E-05	C2	100	25	15	3	1517	1517	70	58	1	Detected	E05_C2
E-05	C3	100	30	15	3	1517	1517	94	82	1	Detected	E05_C3
E-06	C1	90	20	15	5	5308	5108	118	4	7	Detected	E06_C1
E-06	C2	100	25	15	3	5308	5307	828	39	7	Partial	E06_C2
E-07	C1	90	20	15	5	1406	1401	388	310	1	Detected	E07_C1
E-07	C2	100	25	15	3	1406	1405	530	488	1	Detected	E07_C2
E-07	C3	90	20	15	3	1406	1401	388	353	1	Detected	E07_C3

E-08	C1	90	20	15	5	5793	3974	8	0	0	Not detected	E08_C1
E-08	C2	100	25	15	3	5793	5793	18	15	1	Detected	E08_C2
E-08	C3	90	35	15	5	5793	3974	184	27	1	Detected	E08_C3
E-09	C1	90	20	15	5	1450	510	8	0	0	Not detected	E09_C1
E-09	C2	100	25	15	3	1450	1450	48	42	1	Detected	E09_C2
E-09	C3	100	30	15	3	1450	1450	70	60	1	Detected	E09_C3
E-10	C1	90	20	15	5	318	193	12	2	2	False detection	E10_C1
E-10	C2	100	25	15	3	318	318	64	18	2	False detection	E10_C2
E-10	C3	100	30	15	3	318	318	108	40	2	False detection	E10_C3

Table 2. Summary of Outcomes

Item	Transformation, region size	C1 (default)	C2 (moderate)	C3 (image-specific)
E-01	Translation, 2,122 px	Detected	Partial	Detected
E-02	Translation, 9,852 px	Detected	Detected	Detected
E-03	Translation, 403 px	Not detected	False detection	False detection
E-04	Translation, 28,405 px	Detected	Partial	Partial
E-05	Rotation R2, 14,186 px	Detected	Detected	Detected

E-06	Rotation R180, 2,632 px	Detected	Partial	Not run
E-07	Scaling S101, 9,852 px	Detected	Detected	Detected
E-08	Scaling S40, 15,478 px	Not detected	Detected	Detected
E-09	Vertical distortion V115, 14,186 px	Not detected	Detected	Detected
E-10	Combination S80 R45, 3,540 px	False detection	False detection	False detection

Across the ten forged images, the default configuration C1 detected six images (E-01, E-02, E-04, E-05, E-06, E-07), found no cluster in three (E-03, E-08, E-09) and gave a false detection in one (E-10). The relaxed configuration C2 detected five images, gave a partial result in three (E-01, E-04, E-06) and false detections in two (E-03, E-10). The image-specific configuration C3, run on nine images, detected six, gave one partial result (E-04) and two false detections (E-03, E-10). Taking only the unbiased configurations C1 and C2 together, eight of the ten images were detected at least once, and E-03 and E-10 were never detected correctly.

Several patterns can be seen. Copied regions with distinctive texture and no strong geometric transformation (E-01, E-02, E-04, E-05, E-06, E-07) were found at the default settings. Region size alone did not decide the outcome: E-08 (15,478 px) and E-09 (14,186 px) are large but were missed at the default settings because of strong scaling and distortion, while the small regions in E-01 (2,122 px) and E-06 (2,632 px) were found. The small rotation and the 180 degree rotation (E-05, E-06) did not prevent detection, which is consistent with BRISK being a rotation invariant descriptor. The strong downscale (E-08, scale 40%) and the vertical stretch (E-09) were missed at the default settings and found only after matching was relaxed to 25% or more. Relaxing Matching also added lines on repetitive structure that is not part of the forgery, such as the building facade in E-01, the lettering and windows in E-03, the book spines and shelf rows in E-04 and the corrugated roof in E-06. The two hardest cases were E-03 (the smallest region, 403 px) and E-10 (rotation by 45 degrees combined with scaling, in a plain low-texture area with only 318 keypoints): in all parameter sets the lines appeared in places that the ground truth does not mark. A non-zero cluster count is therefore not proof of a correct detection, as E-10 shows, and the Regions value reported by Sherloq is only an estimate. It was 7 for E-06, whose ground truth contains two regions.

5. Reporting

All settings, tool outputs and screenshots were stored per item (E-01 to E-10) and are attached in Appendix A. The procedure can be reproduced by cloning Sherloq at commit 3fe95fc, installing the requirements, selecting the same ten CoMoFoD sets, and running the Copy-Move Forgery tool with the parameters listed in Table 1. The ten forged images were hashed again after all runs to confirm that they were not modified.

Findings

List each significant finding, the evidence it comes from, and why it matters to the case. Clearly separate facts (what was observed) from interpretation (what it may mean).

Task 1

Finding ID	Artifact / data	Source (item and path)	Description	Significance
F-01	Deleted JPEG directory entry	E-02 / FAT32 DF_GROUP3 root	TestDisk identified evidence_group3.jpg as a deleted 61,320-byte file.	Confirms that permanent deletion removed normal directory visibility but did not immediately eliminate recoverable file data.
F-02	Recovered JPEG	03_Recovery\evidence_group3.jpg	TestDisk copied the deleted file successfully with 1 ok and 0 failed.	Demonstrates successful deleted-file recovery from the forensic image.
F-03	SHA-256 of original and recovered JPEG	E-03 / 01_Original and 03_Recovery	Original and recovered hashes are identical: 458A79...F4D591.	Confirms the recovered JPEG is bit-for-bit identical to the ground-truth file.
F-04	SHA-256 of forensic image	E-02 / 02_Acquisition	Image hash before and after TestDisk analysis is identical: C2E427...EB24A1.	Supports integrity of the working forensic image during examination.

Task 2

Finding ID	Artifact / data	Source (item and path)	Description	Significance
F-01	Copy-move result, translation, region 2,122 px (two pedestrians)	E-01, dataset\images\001_F.png	At C1 and C3 the lines connect the two marked pedestrians and nothing else. At C2 the pedestrians are still matched, but additional lines join repeated windows on the building facade. See E01_C1, E01_C2, E01_C3, GT-E01 and Figure 4	The default settings are sufficient. Relaxing Matching adds matches on repetitive architecture that is not part of the forgery
F-02	Copy-move result, translation, region 9,852 px (two plates)	E-02, dataset\images\005_F.png	In all three runs the lines connect the two marked regions, with several hundred matches. See E02_C1, E02_C2, E02_C3, GT-E02	A large textured copied region is detected under every setting tested

F-03	Copy-move result, translation, region 403 px	E-03, dataset\images\010_F.png	At C1 no cluster was found. At C2 and C3 the lines join lettering and window bases of the building, not the two small marked areas. See E03_C1, E03_C2, E03_C3, GT-E03	The tiny region contains too few distinctive keypoints. Relaxed settings produced a false detection on repetitive structure
F-04	Copy-move result, translation, region 28,405 px (books)	E-04, dataset\images\024_F.png	At C1 the clusters lie inside the marked book area. At C2 and C3 six regions were reported, and extra lines run toward the upper right shelf and along the red books. See E04_C1, E04_C2, E04_C3, GT-E04	The marked area is found, but it consists of repeated vertical book spines, so many internal matches exist and relaxed settings add lines outside the ground truth
F-05	Copy-move result, rotation R2, region 14,186 px (pigeon)	E-05, dataset\images\050_F.png	In all three runs the lines connect the two pigeon regions marked in the ground truth. See E05_C1, E05_C2, E05_C3, GT-E05	A small rotation does not hinder detection, and relaxing parameters only increases the number of matches
F-06	Copy-move result, rotation R180, region 2,632 px (roof)	E-06, dataset\images\063_F.png	At C1 the lines are at the marked patches. At C2 additional lines are spread over the roof. Sherloq reported 7 regions, while the ground truth has two. C3 was not run. See E06_C1, E06_C2, GT-E06	A 180 degree rotation is handled at default settings. The roof texture is repetitive, so relaxed settings add unrelated lines, and the Regions value is not a reliable count
F-07	Copy-move result, scaling S101, region 9,852 px (two plates)	E-07, dataset\images\085_F.png	In all three runs the lines connect the two marked regions. See E07_C1, E07_C2, E07_C3, GT-E07	A one percent scaling does not affect detection
F-08	Copy-move result, scaling S40, region 15,478 px	E-08, dataset\images\103_F.png	At C1 no cluster was found. At C2 and C3 the lines join the small lower-left	Strong downscaling reduces matching, so detection was possible only after Matching was relaxed

	reduced to 2,620 px (plants)		region and the larger upper-right region marked in the ground truth. See E08_C1, E08_C2, E08_C3, GT-E08	
F-09	Copy-move result, vertical distortion V115, region 14,186 px (pigeon)	E-09, dataset\images\126_F.png	At C1 no cluster was found. At C2 and C3 the lines connect the two pigeon regions. See E09_C1, E09_C2, E09_C3, GT-E09	The stretched copy alters descriptors enough to be missed at default settings, but it is found with relaxed Matching
F-10	Copy-move result, combination S80 R45, region 3,540 px	E-10, dataset\images\168_F.png	Only 318 keypoints were found. In all three runs the lines lie on the plain lower wall, not on the two marked squares. See E10_C1, E10_C2, E10_C3, GT-E10 and Figure 5	A plain, low-texture region that is both scaled and rotated was not detected, and clusters were reported at wrong locations (false detection)

Timeline of Events

Task 1

Date / time (WIB)	Event	Source artifact	Finding ID
05-Oct-2026 20:32	Prepared FAT32 volume created; System Volume Information timestamp visible in TestDisk	E-01 / DF_GROUP3	
05-Oct-2026 21:14	evidence_group3.jpg present on USB and later permanently deleted	E-03 / file timestamp	F-01
05-Oct-2026 22:06	Raw forensic image files created by FTK Imager	E-02 / acquisition output	
05-Oct-2026, same examination session	Deleted JPEG recovered and hashes verified	E-02 and E-03	F-02 to F-04

Task 2

Date / time (WIB)	Event	Source artifact	Finding ID
2026-10-05 14:07	Sherloq repository placed in the working folder (commit 3fe95fc) and Python 3.11.15 virtual environment created with uv 0.11.23.	sherloq/, sherloq/.venv	-
2026-10-05 15:56	CoMoFoD small v2 dataset obtained (manual download by the examiner) and extracted to the working folder.	comofod_small/	-

2026-10-05 15:59	Dataset assembled: 10 forged images, 10 ground-truth masks and 10 originals copied. manifest.csv written. SHA-256 hashes at acquisition recorded.	dataset/manifest.csv, dataset/hashes_sha256.txt	-
2026-10-05 16:23	First integrity verification: all 30 files matched the acquisition hashes.	Appendix B, hash log	-
2026-10-05 18:40 to 18:42	E-01 run with C1, C2 and C3. C1 and C3 detected the region and C2 gave a partial result.	E01_C1, E01_C2, E01_C3	F-01
2026-10-05 18:43 to 18:44	E-02 run with C1, C2 and C3. All three configurations detected the region.	E02_C1, E02_C2, E02_C3	F-02
2026-10-05 18:54 to 18:56	E-03 (smallest region, 403 px) run with C1, C2 and C3. Not detected at C1, false detection at C2 and C3.	E03_C1, E03_C2, E03_C3	F-03
2026-10-05 18:57 to 18:59	E-04 run with C1, C2 and C3. Detected at C1, partial at C2 and C3.	E04_C1, E04_C2, E04_C3	F-04
2026-10-05 19:00 to 19:02	E-05 run with C1, C2 and C3. All three configurations detected the region.	E05_C1, E05_C2, E05_C3	F-05
2026-10-05 19:03 to 19:04	E-06 run with C1 and C2. Detected at C1, partial at C2. C3 was not run.	E06_C1, E06_C2	F-06
2026-10-05 19:04 to 19:06	E-07 run with C1, C2 and C3. All three configurations detected the region.	E07_C1, E07_C2, E07_C3	F-07
2026-10-05 19:07 to 19:09	E-08 run with C1, C2 and C3. Not detected at C1, detected at C2 and C3.	E08_C1, E08_C2, E08_C3	F-08
2026-10-05 19:10 to 19:11	E-09 run with C1, C2 and C3. Not detected at C1, detected at C2 and C3.	E09_C1, E09_C2, E09_C3	F-09
2026-10-06 09:25 to 09:27	E-10 (combined scaling and rotation) run with C1, C2 and C3. False detection in all three configurations.	E10_C1, E10_C2, E10_C3	F-10
2026-10-06 09:46 to 09:47	Environment evidence captured: Sherloq main window (version 0.94f) and Python 3.11.15, Git commit 3fe95fc and uv 0.11.23	ENV-2, ENV-1	-
2026-10-06 09:48	Second integrity verification after all Sherloq runs: SHA-256 of the ten forged images computed with PowerShell Get-FileHash. All ten values matched the acquisition hashes.	ENV-3	-

2026-10-06	Outcomes classified against the ground-truth masks and summarised in Table 1, and the report was drafted.	Table 1, Analysis	F-01 to F-10

Limitations and Assumptions

Task 1:

- No hardware write blocker was used. After permanent deletion, the workflow minimized intentional writes to the USB and acquisition was performed immediately with FTK Imager. This limitation is explicitly acknowledged.
- The USB is flash media with an internal controller and wear-leveling behavior. The experiment validates recovery from the logical sectors exposed to the operating system and does not claim physical NAND-level acquisition.
- The Windows Disk ID is not a hardware serial number. A hardware serial number was not separately recorded during the exercise.
- FTK Imager built-in verification used MD5 and SHA-1. Because these algorithms are not preferred for modern collision resistance, SHA-256 was additionally calculated for the forensic image and for file-level integrity checks.
- The exercise intentionally used a freshly prepared FAT32 volume and one known target file. Results should not be generalized to overwritten, fragmented, encrypted, TRIM-affected, or physically damaged media.

Task 2:

- The ten images were selected by the examiner, not drawn at random, and the sample is small, so the results show how the tool behaves in these cases and should not be read as a detection rate for the tool in general.
- The outcome classes (Detected, Partial, Not detected, False detection) were assigned by visual comparison of each screenshot with the ground-truth mask and are therefore a judgment of the examiner.
- Configuration C3 was chosen with knowledge of the ground truth. It shows the best result reachable on these images and is not a realistic forensic scenario, so the conclusions rely mainly on C1 and C2.
- No unmodified images were tested. For this reason the false detection rate on genuine images was not measured. In this report, "false detection" means that clusters were found at locations the ground truth does not mark.
- The value of the Regions field is only an estimate made by Sherloq. Only the BRISK detector, one dataset (CoMoFoD small v2, uncompressed PNG) and one Sherloq version (commit 3fe95fc) were used. The ground-truth masks were assumed to be correct.

Conclusion

Task 1

Task 1 was completed successfully. A SanDisk Cruzer Switch USB flash drive was prepared as a FAT32 test medium, a known JPEG file was copied to it, verified, and permanently deleted. FTK Imager produced a physical Raw (dd) forensic image and reported matching MD5 and SHA-1 verification values with no bad blocks. TestDisk 7.2 subsequently identified the deleted evidence_group3.jpg entry and recovered it from the forensic image.

The recovered file had the same SHA-256 value as the retained original file, 458A79BCB8303CB8A077D26699552D3E33434FEBCA0E4293E7F1547D68F4D591, establishing a bit-for-bit match. The forensic image SHA-256 value also remained unchanged before and after examination, C2E427E0A413D0D6A43C9ED750C16FB2A9177ED9784F7D8211DCA699D8EB24A1. The investigative questions were therefore answered affirmatively: the deleted JPEG was recoverable, the recovered output was identical to the original, and the integrity of the forensic image was maintained during analysis.

Task 2

The Copy-Move Forgery tool of Sherloq with the BRISK detector located the cloned regions in six of the ten forged images at its default settings (C1). With the fixed relaxed configuration (C2), two more images (E-08 and E-09) were detected, so eight of ten images were detected by at least one of the two unbiased configurations, while the smallest region (E-03, 403 px) and the combined scaling and rotation case (E-10) were never located correctly. In answer to the second question, regions with distinctive texture and no strong transformation, including small and 180 degree rotations, were detected at the default settings, whereas strong downscaling (S40) and vertical distortion were detected only after relaxing Matching. Relaxing Matching also produced extra lines on repetitive structures (building facades, book spines, roof texture). The trade-off between sensitivity and false detection is therefore the main practical issue when using the tool, and an examiner should not rely on the cluster count alone.

Recommendations

Task 1:

- For future exercises, use a hardware write blocker when available to further reduce the possibility of writes to the original media during acquisition.
- Continue using SHA-256 or a stronger modern hash as the primary integrity record, while retaining tool-generated MD5/SHA-1 values only when produced automatically for compatibility or audit purposes.

Task 2:

Further work could test the ORB and AKAZE detectors, unmodified images to measure false detections, the post-processed variants of the CoMoFoD images, and other datasets such as CASIA v2.0 and the SIFT-based dataset of the University of Udine. Examiners should run more than one parameter set and review the match lines visually against the image content, since a single configuration can miss small or strongly transformed regions or flag unrelated repeated textures.

Appendices

Task 1:

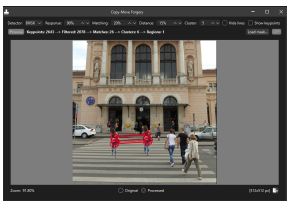
Appendix A. Case Folder Structure

```
DigitalForensics_Assignment1\
├── 01_Original\evidence_group3.jpg
├── 02_Acquisition\Group3_SanDisk16GB_20261005.001
│   ├── Group3_SanDisk16GB_20261005.001.csv
│   └── Group3_SanDisk16GB_20261005.001.txt
├── 03_Recovery\evidence_group3.jpg
├── 04_Screenshots\...
└── 05_Hash\original_sha256.txt / image_sha256_acquisition.txt /
image_sha256_post_analysis.txt
```

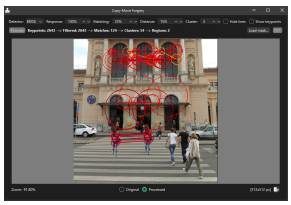
The screenshots are embedded inline in the relevant forensic phase so that each figure remains associated with the action it documents. The FTK-generated directory listing and acquisition text report were retained with the raw image.

Task 2:

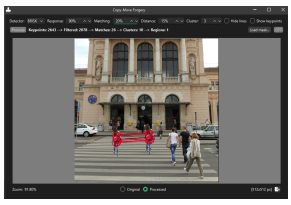
Appendix A: Screenshots



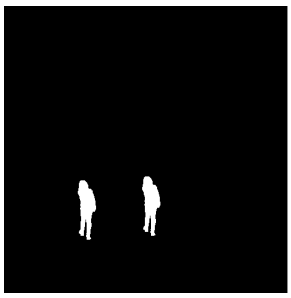
E01_C1



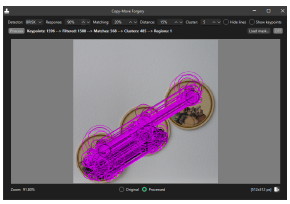
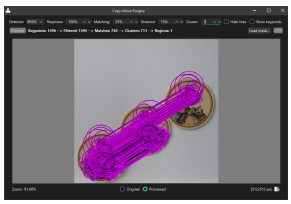
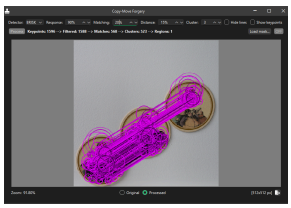
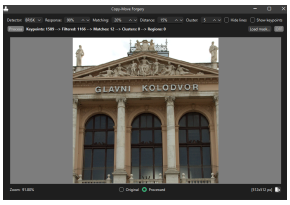
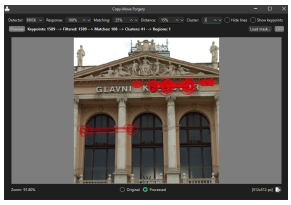
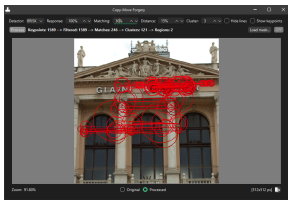
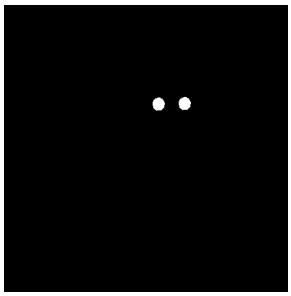
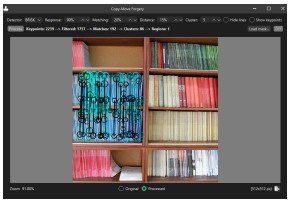
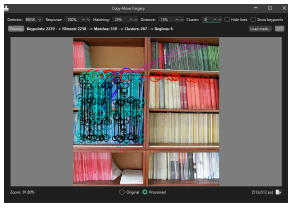
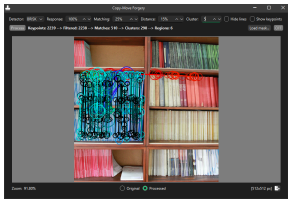
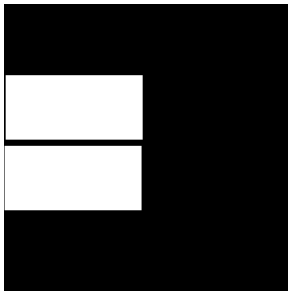
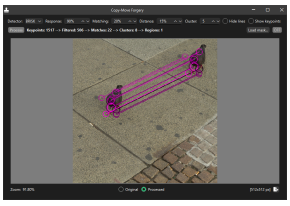
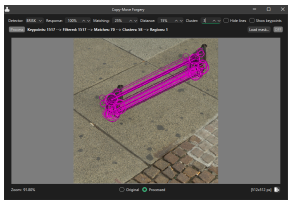
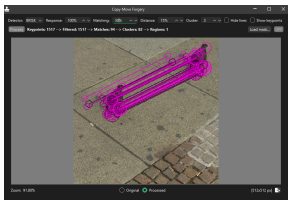
E01_C2

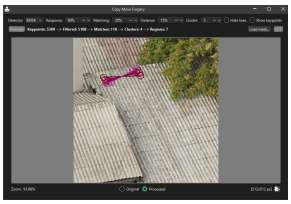
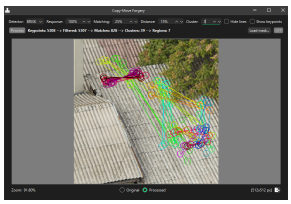
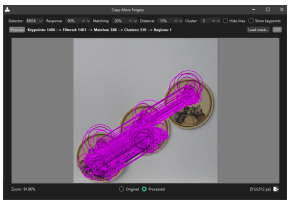
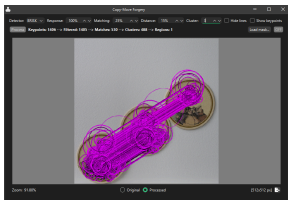
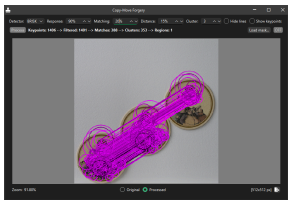
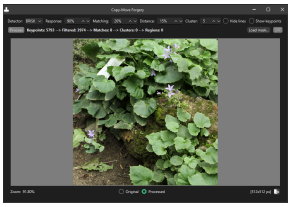
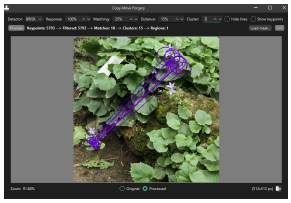
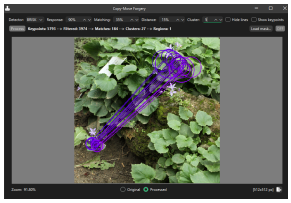
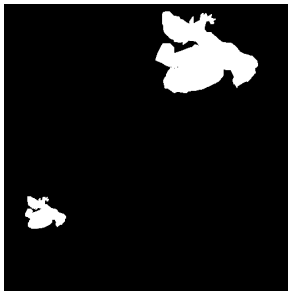
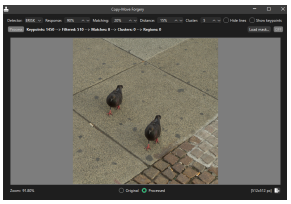
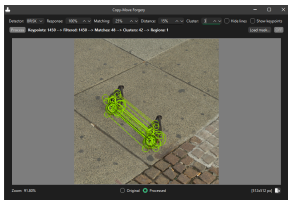
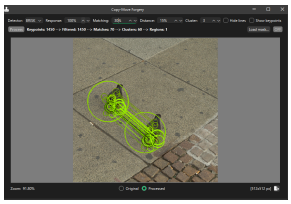


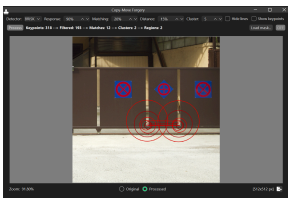
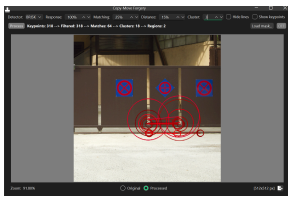
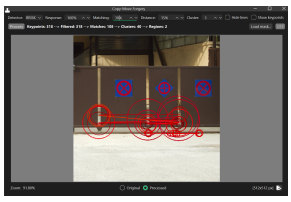
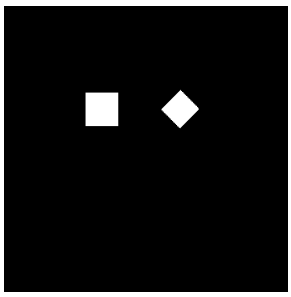
E01_C3



E01_GroundTruth

 E02_C1	 E02_C2	 E02_C3	 E02_GroundTruth
 E03_C1	 E03_C2	 E03_C3	 E03_GroundTruth
 E04_C1	 E04_C2	 E04_C3	 E04_GroundTruth
 E05_C1	 E05_C2	 E05_C3	 E05_GroundTruth

 E06_C1	 E06_C2	Not Runu	 E06_GroundTruth
 E07_C1	 E07_C2	 E07_C3	 E07_GroundTruth
 E08_C1	 E08_C2	 E08_C3	 E08_GroundTruth
 E09_C1	 E09_C2	 E09_C3	 E09_GroundTruth

 E10_C1	 E10_C2	 E10_C3	 E10_GroundTruth

Appendix B: Hash log (SHA-256)

File	SHA-256
images/001_F.png	1006d4449c0df42f7b5537f1b1f53d5190ea802c6225646a8e4f87e2cb825e7c
images/005_F.png	ae95bfbc784b2208d4625758f9b756501915950573629d249db1bb286575d71d
images/010_F.png	f50f55ab56e165af9d18c5d901203e2514986b50c4742013df5b86bf6b995621
images/024_F.png	da6a6faa47a36c5ef24445fc4e131a9ec78f4fd9bcc84b22e7ee3adf67f8e9f0
images/050_F.png	7118c50a7b96a36aa01ca153ea8fe47edd89d28f71cc73c027a34229711a7bdc
images/063_F.png	33f79fc8084c23f79b76feebde7fa41a574806fbe880420d5781fd5e3c763a97
images/085_F.png	8c4ca4ce76f3580212c968e8abf4e78803a9a6b76c86e54913ecaf8d5e317b32
images/103_F.png	0503b4194c021d948cf171d5e617aff360c498adc0edd1ceb1fe797dc31f62e6
images/126_F.png	f6147f4fe285bc16c3b6ded1b5250ac374731bb4430fa7e90cd0a1b80f55f8e5
images/168_F.png	fd421fc7db0b5023adc159be26213d017d6ab130b37c5fc7fb741bb85a39d65b
gt/001_B.png	6756737c9c6f15712af9d379d8e6a701866f94de4f3e52d885ca1b4a09e3f781
gt/005_B.png	801dd0cc281a548dba8d9b58355d0224d56b56478fb3534074a1ac5d6561d852

gt/010_B.png	35f6193ced8737c4314531002dff34bf25f9607f4fc537a88ee8ba6a4549a356
gt/024_B.png	7465e932b3cbf6c09febd4e3c0387ee55ba8f440ec5e7cfc30c0ae0295adf500
gt/050_B.png	e65334febee185f33402ba440ccdc27129688c0a4a826355fd8fac2b6b3cfe8c
gt/063_B.png	713c5a971ef75ee0109c8d68476b3ca5ff1dd55367e4e6f230a9980848e440fb
gt/085_B.png	9bb60a5645221a436a6f63129ec57d1794527d467e63c5062ad7f6ef65ae0e80
gt/103_B.png	5f8ac1736304d1514d85d559b75fa45e71e41e4258b0262c6c01ce4e9c309ace
gt/126_B.png	d72d9512dda4c93749b80ab329332eb9cd0275afeb4723f987eefa3b622aa7c5
gt/168_B.png	a5305d01222325441f8ab452c124882c7dded10130cb6e157496ad44b92354a1

References

[1] Kent, K., Chevalier, S., Grance, T., and Dang, H. (2006). Guide to Integrating Forensic Techniques into Incident Response, NIST Special Publication 800-86.

[2] Brezinski, D. and Killalea, T. (2002). Guidelines for Evidence Collection and Archiving, IETF RFC 3227.

[3] ISO/IEC 27037:2012, Guidelines for identification, collection, acquisition and preservation of digital evidence.

[4] NIST Computer Forensics Tool Testing (CFTT) Program, https://www.nist.gov/itl/ssd/software-quality-group/computer-forensics-tool-testing-program-cftt

[5] Bartoli, G. Sherloq: an open-source digital image forensic toolset, https://github.com/GuidoBartoli/sherloq

[6] Tralic, D., Zupancic, I., Grgic, S., and Grgic, M. (2013). CoMoFoD: New database for copy-move forgery detection. Proceedings of ELMAR 2013, https://www.vcl.fer.hr/comofod/

[7] Leutenegger, S., Chli, M., and Siegwart, R. (2011). BRISK: Binary Robust Invariant Scalable Keypoints. Proceedings of ICCV 2011.